1.2 Atoms, Molecules & Stoichiometry

Question Paper

Course	CIE A Level Chemistry
Section	1. Physical Chemistry
Topic	1.2 Atoms, Molecules & Stoichiometry
Difficulty	Easy

Time allowed: 20

Score: /10

Percentage: /100

Question 1

The atomic mass is a way of representing the mass of an atom.

Which of the following definitions is correct for the term atomic mass?

A. The mass of an electron

B. 1/12 the mass of carbon-12 atom

C. The mass of hydrogen-1 atom

D. The mass of a proton

[1 mark]

Question 2

Elements can be found to have different isotopes.

What is different about the nuclei of isotopes?

A. The same number of protons, but different number of neutrons

B. The same number of protons, and the same number of neutrons

C. A different number of protons, and a different number of neutrons

D. A different number of protons, and the same number of neutrons

[1 mark]

Question 3

Which of the following show the relative formula mass of ammonium sulfate, $(NH_4)_2SO_4$?

Use of the data booklet is relevant to this question.

A. 98.1

B. 98

C. 132.1

D. 132

[1 mark]

Question 4

Iron has many different naturally occurring isotopes.

Isotope	% Abundance	Atomic mass
^{54}Fe	5.845	53.9396
^{56}Fe	91.754	55.9349
^{57}Fe	2.119	56.9354
^{58}Fe	0.282	57.9333

Using the information given, what is the average atomic mass of iron?

A. 56.1858 amu

B. 55.500 amu

C. 59.270 amu

D. 55.845 amu

[1 mark]

Question 5

Use of the data booklet is relevant to this question

What is the number of molecules in 500 cm^3 of oxygen at standard temperature and pressure?

A. 3.00×10^{23}

B. 3.00×10^{26}

C. 1.25×10^{21}

D. 1.34×10^{22}

[1 mark]

Question 6

When the following equation is balanced correctly, using the smallest whole number coefficients, which row represents the coefficients?

$$\_\_Mg_3N_2(s) + \_\_H_2O(l) \rightarrow \_\_Mg(OH)_2(aq) + \_\_NH_3(aq)$$

	Mg_3N_2	H_2O	$Mg(OH)_2$	NH_3
A	1	6	3	2
B	1	3	3	1
C	2	6	2	2
D	2	6	3	2

[1 mark]

Question 7

Which one of the following is the correct net ionic equation for the reaction (if one occurs) between $CaCl_2$ and $AgNO_3$?

A. $Ca^{2+}(aq) + 2AgNO_3(aq) \rightarrow 2Ag^{+}(s) + Ca(NO_3)_2(aq)$

B. $CaCl_2(aq) + 2Ag^{+}(aq) \rightarrow 2AgCl(s) + Ca^{2+}(aq)$

C. $Ca^{2+}(aq) + 2NO_3^{-}(aq) \rightarrow Ca(NO_3)_2(aq)$

D. $Ag^{+}(aq) + Cl^{-}(aq) \rightarrow AgCl(s)$

[1 mark]

Question 8

Which one of the following is the correct net ionic equation for the reaction (if one occurs) between $NaC_2H_3O_2(aq)$ and $HCl(aq)$?

A. $C_2H_3O_2^{-}(aq) + HCl(aq) \rightarrow CCl^{-}(aq) + 2H_2(aq) + CO_2(aq)$

B. $C_2H_3O_2^{-}(aq) + H^{+}(aq) \rightarrow HC_2H_3O_2(aq)$

C. $Na^{+}(aq) + Cl^{-}(aq) \rightarrow NaCl(aq)$

D. $NaC_2H_3O_2(aq) + H^{+}(aq) \rightarrow HC_2H_3O_2(aq) + Na^{+}(aq)$

[1 mark]

Question 9

Use of the data booklet is relevant to this question.

A compound was found to be 52.2% by weight carbon, 13.0% hydrogen and 34.8% oxygen.

What is the empirical formula of this compound?

A. $C_2H_4O_2$

B. $C_4H_{13}O_2$

C. C_2H_6O

D. C_6H_2O

[1 mark]

Question 10

Use of the data booklet is relevant to this question.

A sample of hydrated calcium sulfate, $CaSO_4 \bullet xH_2O$, has a relative formula mass of 172.2

What is the value of x?

A. 1

B. 2

C. 3

D. 4

[1 mark]

Model Answers: Easy

1

The correct answer is **B** because:

- Relative atomic mass is the mass of one element compared to another.
- This is compared to carbon-12 as the standard of reference.
- This is defined as the **atomic mass unit** (amu) which has the value of 1.6606 x 10^{-24}g.
- This takes into account the mass of the different **isotopes.**
- Relative atomic mass of element E = $\dfrac{\textit{average mass of one atom of E}}{\frac{1}{12}\ \textit{the mass of one atom of Carbon 12}}$

A is incorrect as the mass of the electron is 1/2000 the mass of a proton or neutron and a very small fraction of the overall mass of the atom.

C is incorrect as this is the way the relative atomic mass was calculated, but because of **isotopes** and the role of carbon in organic compounds it was changed to carbon-12, ^{12}C.

D is incorrect as the atom will contain neutrons which also have a mass.

2

The correct answer is **A** because:

- The **atomic mass** of an atom is determined by the number of protons and neutrons in the **nucleus.**
- An **isotope** is an atom that has a different number of neutrons but still has the same chemical properties.

B is incorrect as if the atom has the same number of **neutrons** and **protons** then it will not be an isotope.

C & **D** are incorrect as if the atom has a different number of **protons** then it will be a different atom.

3

The correct answer is **C** because:

- The term **molecular mass** is not used when referring to **ionic** compounds such as ammonium sulfate.
- The **relative formula mass** of ionic and giant covalent compounds is calculated based on the **empirical formula.**
- The relative formula mass is calculated by adding the relative atomic masses of all the atoms present.
- $M_r = 2A_r(N) + 8A_r(H) + A_r(S) + 4A_r(O)$

  $= 28 + 8 + 32.1 + 64$

  $= 132.1$

A is incorrect as this answer has not multiplied all the atoms inside the brackets by the number outside the bracket for ammonium.

B is incorrect as this has not used the correct atomic mass for sulfur and not multiplied all the atoms inside the brackets by the number outside the bracket for ammonium.

D is incorrect as this has not used the correct atomic mass for sulfur.

4

The correct answer is **D** because:

- Atoms of the same element with a different number of neutrons are called **isotopes.**
- To calculate the average atomic mass of an element the **percentage abundance** is taken into account.

1. Multiply the atomic mass by the percentage abundance for each isotope.
2. Add all values from step 1 together.
3. Divide by 100 to get average relative atomic mass.

    - $A_r(Fe) = \frac{(5.845 \ X \ 53.9396) + (91.754 \ X \ 55.9349) + (2.119 \ X \ 56.9354) + (0.282 \ X \ 57.9333)}{100} =$ **55.845 amu**
    - When doing calculations make sure the number of decimal places is the same as the original data given.

A is incorrect as this answer is an average of the masses and does not take into account the percentage abundance.

B is incorrect as this has taken the relative atomic mass from the periodic table.

C is incorrect as this is an incorrect calculation.

5

The correct answer is **D** because:

- The **Avogadro constant,** symbol ***L***, is the number of particles (atoms, molecules or ions) in a mole. The approximate value of ***L =*** 6.02×10^{23} mol^{-1}.
- The relationship between the number of moles in a sample and the number of particles it contain is as follows:
    - *number of particles =* ***L*** *x number of moles* or *n = Ln*
- 500 cm^3 is the same as 0.5 dm^3.
- At standard temperature and pressure (s.t.p) 22.4 dm^3 of a gas = 6.02×10^{23} molecules.
- Therefore, the number of particles = $\frac{0.5}{22.4} \times (6.02 \times 10^{23}) = 1.34 \times 10^{22}$

A is incorrect as this answer is $6.0 \times 10^{23} \times 0.5$.

B is incorrect as the units have not been converted to dm^3, giving the answer $6.0 \times 10^{23} \times 500$.

C is incorrect as this answer is $6.0 \times 10^{23} / 500$.

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C is incorrect as this answer is $6.0 \times 10^{23} / 500$.

6

The correct answer is **A** because:

Step		Working out
1	The number of each atom on either side of the reaction **without coefficients** is shown.	Mg_3N_2 (s) + H_2O (l) → $Mg(OH)_2$ (aq) + NH_3 (aq) Mg x 3 Mg x 1 N x 2 N x 1 H x 2 H x 5 O x 1 O x 2
		Mg_3N_2 (s) + **6**H_2O (l) → **3**$Mg(OH)_2$ (aq) + **2**NH_3

			(aq)
2	There are **equal** numbers of H but **unequal** numbers of Mg, N and O on either side. Change the numbers to match by adjusting the **coefficients**.	Mg x 3 N x 2 H x 12 O x 6	Mg x 3 N x 2 H x 12 O x 6

B, C & **D** are incorrect as these options do not provide the correct balanced equation.

7

The correct answer is **D** because:

- Ionic equations do not include the spectator ions. Spectator ions are ions that are present in the reaction but do not take part in it.
- In this example, the spectator ions are Ca^{2+} and NO_3^{-}.
- Steps to write the net ionic equation:
  1. Write balanced **molecular equation.**
  2. Write the state symbols (s, l, g, aq) for each substance.
  3. Split strong electrolytes into ions (the **complete ionic equation**).
  4. Cross out the spectator ions that appear on both sides of the complete equation.
  5. Write the remaining substance as the **net ionic equation.**

A, B & **C** are incorrect as they all show the spectator ions $Ca(NO_3)_2$ within the ionic equation.

8

The correct answer is **B** because:

- Ionic equations do not include the spectator ions. Spectator ions are ions that are present in the reaction but do not take part in it.
- Acetic acid is a weak acid so is written as dissociated molecular form $HC_2H_3O_2$ (aq).

- Steps to write the net ionic equation:
  1. Write balanced **molecular equation.**
  2. Write the state symbols (s, l, g, aq) for each substance.
  3. Split strong electrolytes into ions (the **complete ionic equation**).
  4. Cross out the spectator ions on both sides of the complete equation.
  5. Write the remaining substance as the **net ionic equation.**

A is incorrect as the hydrochloric acid would fully dissociate into H^+ and Cl^-.

C & **D** are incorrect as they show the spectator ions.

9

The correct answer is **C** because:

	Steps	Calculation
1	Assume total mass is 100 g, so the mass of each element is the percentage given in the question.	52.2 g carbon 13.0 g Hydrogen 34.8 g Oxygen
2	Convert each **mass to moles** using the molar mass from periodic table.	Moles carbon = 52.2 / 12 = 4.35 Moles hydrogen = 13 Moles oxygen = 2.18
3	Divide by the smallest number.	4.35 / 2.18 = 1.99 13 / 2.18 = 5.96 2.18 / 2.18 = 1
4	Round up to nearest whole number to give ratio.	2:6:1
5	Write out **empirical formula.**	**C_2H_6O**

A, B & **D** are incorrect as they incorrectly calculate the empirical formula.

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The correct answer is **B** because:

- Ionic equations do not include the spectator ions. Spectator ions are ions that are present in the reaction but do not take part in it.
- Acetic acid is a weak acid so is written as dissociated molecular form $HC_2H_3O_2$ (aq).
- Steps to write the net ionic equation:
  1. Write balanced **molecular equation.**
  2. Write the state symbols (s, l, g, aq) for each substance.
  3. Split strong electrolytes into ions (the **complete ionic equation**).
  4. Cross out the spectator ions on both sides of the complete equation.
  5. Write the remaining substance as the **net ionic equation.**

A is incorrect as the hydrochloric acid would fully dissociate into H^+ and Cl^-.

C & **D** are incorrect as they show the spectator ions.

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The correct answer is **C** because:

	Steps	**Calculation**
1	Assume total mass is 100 g, so the mass of each element is the percentage given in the question.	52.2 g carbon 13.0 g Hydrogen 34.8 g Oxygen
2	Convert each **mass to moles** using the molar mass from periodic table.	Moles carbon = 52.2 / 12 = 4.35 Moles hydrogen = 13 Moles oxygen = 2.18

3	Divide by the smallest number.	4.35 / 2.18 = 1.99 13 / 2.18 = 5.96 2.18 / 2.18 = 1
4	Round up to nearest whole number to give ratio.	2:6:1
5	Write out **empirical formula.**	$\mathbf{C_2H_6O}$

A, B & **D** are incorrect as they incorrectly calculate the empirical formula.

10

The correct answer is **B** because:

Steps		Calculation
1	Calculate the **formula mass** for each compound.	$CaSO_4$ = 40.1 + 32.1 + (4 x 16.0) = 136.2 H_2O = 2.0 + 16 = 18
2	There are x molecules of water in the molecule.	136.2 + 18x = 172.2
3	Minus the mass of the calcium sulfate from the total **formula mass.**	172.2 – 136.2 = 36.0
4	Divide by the **molecular mass** of water to find out how many water molecules are present.	36 / 18 = **2**